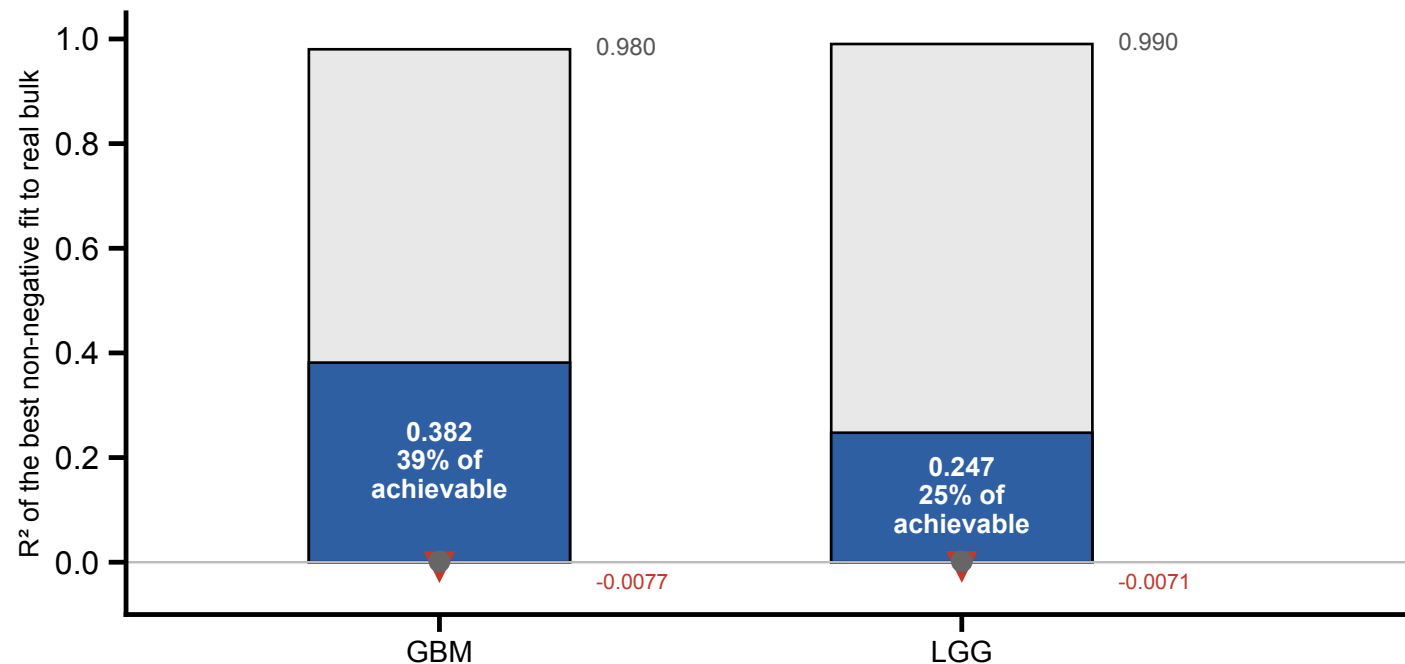


**The mixing model explains a minority of real bulk — but far more than chance, so the reference carries real structure**



floor: gene  $\rightarrow$  cell-type labels shuffled



floor: random non-negative basis



ceiling: best possible 8-dim basis



the reference actually used